

Armaan Singh Chahal

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Education

BSc. Computing Science, Statistics minor, *Simon Fraser University*

(Expected Graduation: May 2027)

Relevant Coursework: DSA, Database Systems, Machine Learning, Software Engineering

GPA: 3.47 (3.75 AI major), *Dean's Honour Roll*

Work Experience

AI Automation Engineer, *BCI, Canada.*

Jan 2026 – Present (ends Apr 2026)

- Supporting Microsoft 365 adoption across 10+ departments by translating technical updates into clear end-user friendly visuals.
- Prototyping an M365 update triage workflow (role-based impact summaries & stakeholder routing) to reduce manual review.
- Packaging and publishing org-wide automation code to Azure DevOps (repo standards, documentation, and release-ready handoff) to support scalable rollout.

Research Assistant – Federated Learning, *Wilfrid Laurier University*

Sept 2025 – Dec 2025

- Researching privacy-preserving federated learning (FedAvg, ADMM) for multi-agent cooperative localization.
- Reviewing literature on FL applications in transportation, digital twins, and continual learning.

Back-End Web Developer, *CJSF 90.1 FM, Burnaby, BC*

Sept 2024 – Dec 2024

- Developed backend with Node.js and Directus CMS, integrated with React frontend
- Designed RESTful APIs using Node.js and PostgreSQL; optimized queries to cut average response time by 30%.
- Added role-based authentication and improved CMS efficiency by 40%

Projects

AI Clinic (PyTorch, FastAPI, Expo, AWS EC2, Supabase) ([link to website](#)) ([github](#))

Jan 2026

- Trained two CNN image classifiers for brain MRI and chest X-ray classification; achieved 99.5% and 94.7% accuracy
- Deployed both .pth models as FastAPI inference services on AWS EC2, exposing REST endpoints for real-time image upload and prediction (/predict/brain, /predict/lung)
- Built an Expo (React Native) TypeScript mobile app with Supabase for authentication and return predicted class
- Implemented input validation, request logging, and model versioning; tracked latency/error rates for deployed endpoints.

PyTorch: TVM Compilation and Optimization Pipeline

Sept 2025 - Nov 2025

- Exported PyTorch models (FCL subnets, ResNet-18) to ONNX and verified output parity with PyTorch.
- Compiled ONNX models to TVM (LLVM target) and benchmarked Runtime for baseline latency and throughput.
- Logged p50/p95 latency and memory usage to prepare for tuning and quantization in upcoming phases.

Skin Lesion Classification AI (PyTorch, ResNet-50, AWS) ([link to website](#))

Feb 2025 – April 2025

- Led a team of 5 to train **ResNet-50** CNN to classify dermatoscopic images with over **90% accuracy**
- Automated preprocessing, data labeling, and model evaluation pipelines
- Created dashboards using Tableau to visualize model performance metrics

Social Media MERN Stack App (MERN stack)

Jan 2025 – Mar 2025

- Built **full-stack** app with chat and JWT authentication
- Designed secure **REST APIs** and role-based access control with MongoDB logging

Technical Skills & Tools

Languages & Databases: Python, Java, JavaScript/TypeScript, SQL, C/C++, Bash, PostgreSQL, MySQL, MongoDB, Haskell

Frameworks & Libraries: React, Angular, Node.js, Express.js, Flask, Bootstrap, Tailwind

Cloud & DevOps: AWS (EC2, S3, SageMaker), Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD

Testing & Debugging: JUnit, pytest, Postman, Chrome DevTools (unit & integration testing)

Software Engineering: OOD, design patterns, Git, Agile/Scrum, RESTful API design, microservices

AI & Data Science: TensorFlow, PyTorch, Pandas, NumPy, Matplotlib; ML (CNNs, supervised & unsupervised learning, deployment)

Certifications:

- AWS: Cloud Practitioner, AI Practitioner, Certified Machine Learning Engineer (ongoing)
- Azure: AZ-900 fundamentals, AZ-104 (ongoing)
- AI Specialization: AutoML, PyCaret, GANs, NLP

Volunteer and Leadership:

Peer Tutor — Calculus & Chemistry, *Simon Fraser University*

September 2023 – April 2024

- Tutored a class of 35 students in Calculus & Chemistry through one-on-one and group sessions, improving comprehension and exam performance while strengthening communication and mentorship skills.